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WEEK :-1

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Dataset Design & Quality Expectations

1. Expected Dataset Structure

The dataset is expected to contain battery charging observations collected over time. A typical row should represent one observation from a charging session.

2. Expected Columns

- Time / Timestamp:-Shows when the battery measurement was recorded.
- Voltage:- Battery voltage measured during charging.
- Current:-Charging current measured at the particular observation.
- Power:-Charging power. If it is not directly available, it may potentially be calculated from voltage and current.
- Temperature:-Battery temperature during charging.

- SOC:-State of Charge of the battery.
- Charging Status:-Indicates the charging condition or charging phase.
- Remaining Charging Time:-Represents the amount of time remaining until the defined charging endpoint.

The exact column names and units will be taken from the actual supplied dataset.

3. Important Metadata

The following information should be recorded:

- Battery type or chemistry
- Battery capacity, if available
- Voltage unit
- Current unit
- Power unit
- Temperature unit
- Time unit
- Sampling interval
- Number of charging sessions
- Charging endpoint
- Missing-value representation

4. Missing Data Checks

The following should be checked:

- Number of missing values in every column
- Percentage of missing values
- Missing values within charging sessions
- Whether missing values occur randomly or in particular periods
- Whether target variables contain missing values

5. Invalid Data Checks

The following values should be investigated:

- Invalid voltage values
- Invalid current values
- Invalid temperature values
- SOC values outside the expected range
- Negative Remaining Charging Time
- Impossible timestamps
- Unexpected charging states

6. Duplicate Checks

Check for:

- Completely duplicated rows
- Duplicate timestamps
- Duplicate observations within the same charging session
- Repeated session
- Repeated measurements caused by data collection problems

7. Target Coverage

1 For SOC:

- Minimum SOC
- Maximum SOC
- Distribution of SOC values
- Number of observations at different SOC levels

2 For Remaining Charging Time:

- Minimum RCT
- Maximum RCT
- Distribution of RCT values
- Number of observations at different charging stages

8. Temporal Risks

The following should be checked:

- Whether timestamps are correctly ordered
- Whether there are gaps in measurements
- Whether charging sessions overlap
- Whether multiple observations have the same timestamp
- Whether the sampling interval is consistent

9. Data Leakage Risks:- Data leakage occurs when information that would not actually be available at prediction time is used by the model.

Possible leakage sources include:

- Future measurements
- Future SOC values
- Final charging time
- End-of-session information
- Features calculated using future observations
- Having the same charging session in both training and testing sets

To reduce leakage, session/cycle-aware or time-aware splitting should be used where possible.

10. Recommended Data Split

- Training data:
- Approximately 70%
- Validation data:
- Approximately 15%
- Testing data:
- Approximately 15%

The exact split may be adjusted depending on the dataset size.
The same charging session should not appear in multiple splits.

11. Dataset Risk Checklist

- Raw dataset preserved
- Missing values checked
- Invalid values checked
- Duplicate rows checked
- SOC range checked

- RCT range checked
- Timestamp structure checked
- Leakage risks checked
- Train/validation/test strategy documented